

Year 8 Science

Sedimentary Rocks - Weathering

Sediments are particles of large rocks – including mountains, hills and cliffs – that have been **weathered** and eroded over time. There are several ways in which weathering occurs: chemical weathering, physical weathering (onion-skin/exfoliation and freeze-thaw) and biological weathering.

Your task:

1. Use the websites listed to find out: **what** each type is, and **where** they are likely to occur.

http://www.geolsoc.org.uk/ks3/gsl/education/resources/rockcycle/page3461.html

- Use the underlined hyperlinks to further explore the website.
- Watch the freeze-thaw animation for a better example of this slow process

http://www.bbc.co.uk/bitesize/ks3/science/environment_earth_universe/rock_cycle/revision/6/

Chemical Weathering

1. In a few sentences, describe how chemical weathering of rocks occurs.

   _____

   _____

   _____

   _____

2. In which climates is chemical weathering most likely to occur?

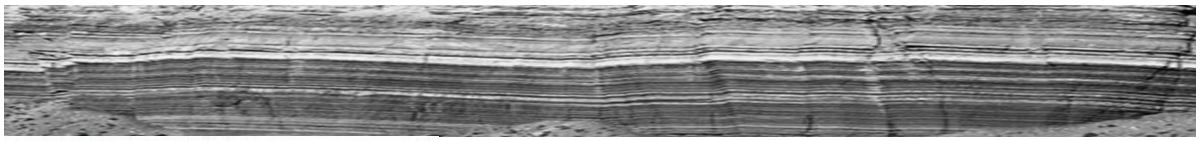
   Hot/dry Cold/icy Humid/rainy

   Give reason for your answer.

   _____

   _____

3. The picture below shows the effects of chemical weathering. Label or caption this picture to show how this statue has been affected by chemical weathering.

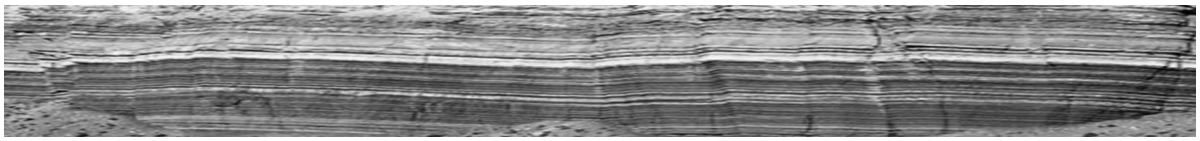




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Present



Physical weathering – Freeze-thaw

1. In a few sentences, list the steps of freeze-thaw weathering. Use the diagram below as a guide

   _____

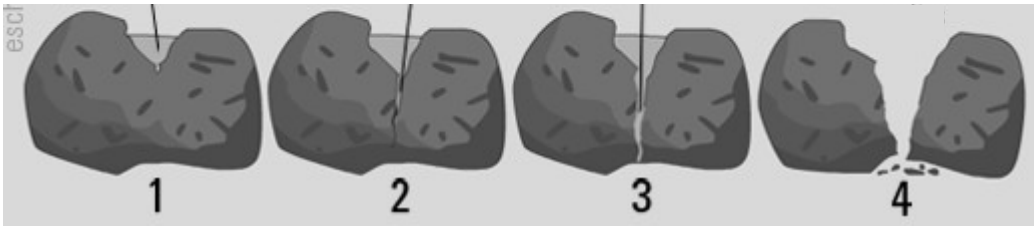
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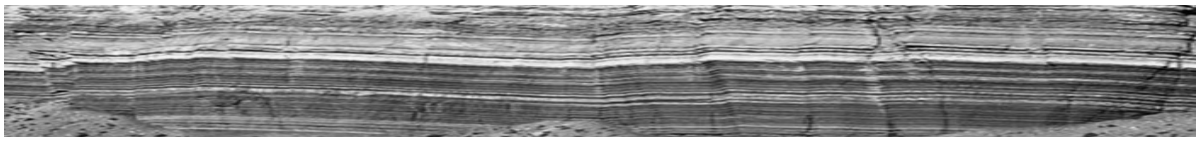
2. In which climates is freeze-thaw weathering most likely to occur?

   Hot/dry Cold/icy Humid/rainy

   Give reason for your answer.

   _____

   _____



Physical weathering – exfoliation

1. In a few sentences, list the steps of exfoliation. Use the diagram below as a guide.

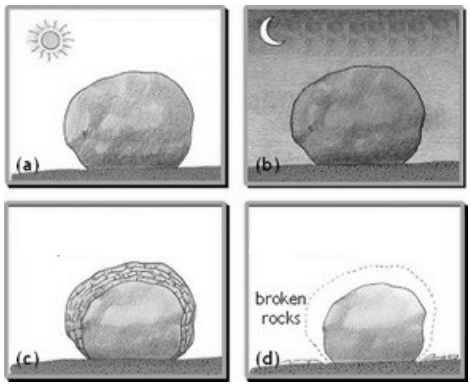
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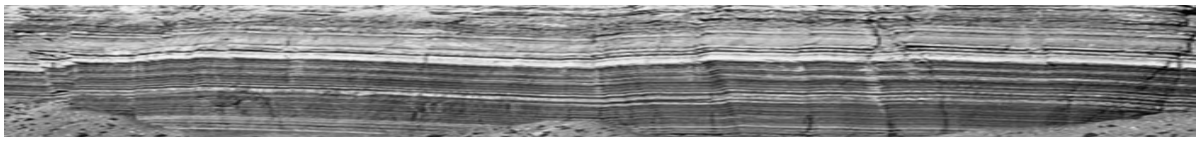
2. In which climates is exfoliation most likely to occur?

   Hot/dry Cold/icy Humid/rainy

   Give reason for your answer.

   _____

   _____



Biological weathering

1. In a few sentences, describe biological weathering

   _____

   _____

   _____

   _____

2. In which climates is biological weathering most likely to occur?

   Hot/dry Cold/icy Humid/rainy

   Give reason for your answer.

   _____

   _____

3. Draw and label a picture, or a set of pictures, below to show biological weathering. Remember, your pictures should show how large rocks are broken into small particles.

